Hong Kong Student Second-hand PIM

COMP3211 Software Engineering | Fall 2026

- Command-line PIM for managing campus second-hand trades
- Presenter names and student IDs: fill before recording

Scope and user value

- A local single-user tool for listing notes, contacts, tasks, and handover events
- The approved workflow covers create, search, update, print, delete, save, and load
- Payment and physical handover remain outside the program

Three requirements

- Creation: create text, task, event, and contact PIRs with validation
- Criterion: combine type, text, and time predicates with !, &&, ||, and parentheses
- Search: return stable, detailed records and treat non-applicable fields as non-matches

Overall design: MVC

- Model owns PIR data, validation, criteria, repository, and .pim persistence
- Controller coordinates parsed commands and model operations
- View renders prompts, records, results, and errors

Example campus trade

- Create a listing: create text content=...
- Create buyer contact, payment task, and campus handover event
- Search: type = text && content contains "ipad"
- Update status to sold, print all records, save campus_market.pim

Verification evidence

- 31 automated tests pass with 0 failures and 0 errors
- 93.6% executable source line coverage: 781 of 834 lines
- US1-US11 each map to executable tests in the requirements coverage report
- Python standard library only

Lesson learned

- Requirements become testable when command grammar and acceptance criteria are explicit
- A small controller boundary makes the model testable without terminal I/O
- Atomic updates and validated persistence protect the user's trade records

Thank you

Questions

- Before recording: replace presenter placeholders, show each member's face and student ID at the beginning, and keep every presenter visible while speaking